Advanced Causal Inference

Lecture 13: Mediation and Treatment Decomposition

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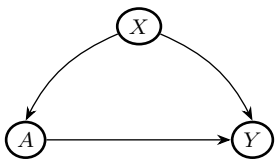
This lecture note is partially based on the lecture notes/textbook of Ying Zhou, Thomas Richardson [Link]

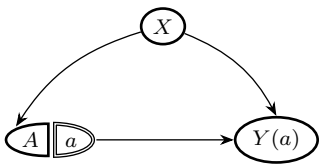
Outline

1. Overview
2. Principal Stratification
3. Pearl's Natural Direct/Indirect Effects
4. Robins and Greenland's Controlled Direct Effect
5. Treatment Decomposition Approach
6. Summary
7. Assignment

By So Far...

- We have focused on a simple "triangular" causal model

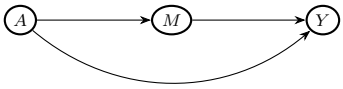


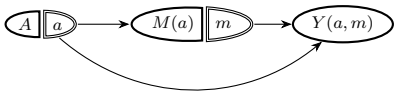


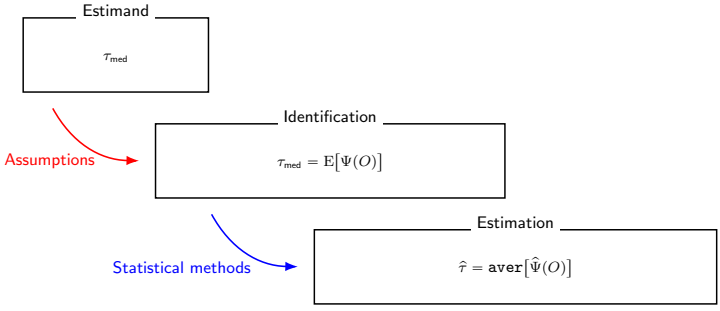
- Causal estimands
  - ATE: $\mathrm{E}\{Y(1) - Y(0)\}$
  - ATT: $\mathrm{E}\{Y(1) - Y(0) | A = 1\}$
  - CATE: $\mathrm{E}\{Y(1) - Y(0) | X\}$
  - Subgroup ATE: $\mathrm{E}\{Y(1) - Y(0) | \mathcal{S}(X)\}$ where $\mathcal{S}(X) \in \{1, \ldots, S\}$

Mediation

- Now, we turn to more complex causal settings
- Our first extension is to consider mediation models





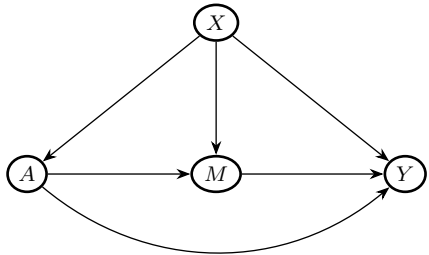


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Post-treatment Variables

- Mediation settings involve post-treatment variables



Motivating Examples

Example (Noncompliance)

In randomized experiments with noncompliance, we can use M to represent the treatment received, which is affected by the treatment assignment A and affects the outcome Y.

Example (Truncation by Death)

In randomized experiments to patients with severe diseases, some patients may die before the measurement of the outcome Y, e.g., the quality of life. The post-treatment variable M in this example is the binary indicator of the survival status.

Motivating Examples

Example (Unemployment)

In job training programs, units are randomly assigned to treatment and control groups, and report their employment status M and wage Y. Then the post-treatment variable is the binary indicator of the employment status M.

Example (Surrogate Endpoint)

In clinical trials, the outcomes of interest (e.g., 30 years of survival) require a long and costly follow-up. Practitioners instead collect data on some other variables early in the follow-up that are easy to measure. These variables are called the **surrogate endpoints**. A concrete example is from clinical trials on HIV patients, where the candidate surrogate endpoint M is the CD4 cell count (recall that CD4 cells are white blood cells that fight infection).

Conditioning on the Post-Treatment Variable

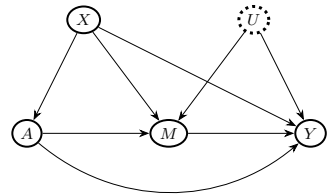
- A naive method is to condition on the observed value of M as if it were a pretreatment covariate
- However, M is fundamentally different from X:
  - M is affected by the treatment, whereas X is not
- It is a common "rule of thumb" that data analyzers should not condition on any post-treatment variables when evaluating the average causal effect of the treatment on the outcome (Cochran, 1957; Rosenbaum, 1984)

Comparing Outcome Distributions

- Let $M(a)$ and $Y(a)$ be the potential mediator and outcome under $A = a$

Assumption (No Unmeasured Confounding between A and (M, Y))

$$A \perp\!\!\!\perp \{M(1), M(0), Y(1), Y(0)\} \,|\, X$$



- Note that there may be unmeasured confounding between M and Y

- Conditioning on $X = x, M = m$, we compare the probabilities:

$$\Pr(Y \mid A = 1, M = m, X = x) \quad \text{and} \quad \Pr(Y \mid A = 0, M = m, X = x)$$

- It measures the difference in outcome distributions in $A = 1$ vs. $A = 0$, given the same value of (M, X)
- However, this interpretation is problematic when M is a post-treatment variable

Rewriting the Probabilities

- Under the Assumption $A \perp\!\!\!\perp \{M(1), M(0), Y(1), Y(0)\} \,|\, X$, we can re-write the probabilities as:

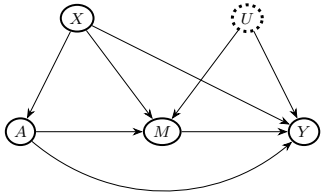
$$\begin{aligned} \Pr(Y \mid A = 1, M = m, X = x) &= \Pr\{Y(1) \mid A = 1, M(1) = m, X = x\} \\ &= \Pr\{Y(1) \mid M(1) = m, X = x\} \end{aligned}$$

$$\begin{aligned} \Pr(Y \mid A = 0, M = m, X = x) &= \Pr\{Y(0) \mid A = 0, M(0) = m, X = x\} \\ &= \Pr\{Y(0) \mid M(0) = m, X = x\} \end{aligned}$$

- Therefore, comparing the two probabilities is equivalent to comparing distributions of $Y(1)$ and $Y(0)$ for different subsets of units

Interpretation Problem

- This comparison does not have a causal interpretation unless $M(1) = M(0)$
- If A affects M, then the units with $M(1) = m$ are different from those with $M(0) = m$
- Conditioning on M introduces "collider bias" which can lead to a biased interpretation of the causal effect



- Conditioning on a collider M while not conditioning on a confounder U opens the path $A \rightarrow \boxed{M} \leftarrow U \rightarrow Y$

Example 1: Noncompliance

Example (Noncompliance)

In randomized experiments with noncompliance, we can use M to represent the treatment received, which is affected by the treatment assignment A and affects the outcome Y.

- Under monotonicity (will discuss this soon)

- $\Pr(Y \mid A = 1, M = 1, X = x)$ $= Y(1)$ for compliers & always-takers

  $\Pr(Y \mid A = 0, M = 1, X = x)$ $= Y(0)$ for always-takers

Example 2: Truncation by Death

Example (Truncation by Death)

In randomized experiments to patients with severe diseases, some patients may die before the measurement of the outcome Y, e.g., the quality of life. The post-treatment variable M in this example is the binary indicator of the survival status.

- If the treatment improves survival status, it could save weaker patients compared to the control
- In this case:
  - Units with $M(1) = 1$ (survival) may be weaker than units with $M(0) = 1$
  - A naive comparison of treated versus control units with $M = 1$ can give biased results favoring the control group

Conditioning on the Joint Potential Values of M

- Frangakis and Rubin (2002) proposed conditioning on the joint potential value of the post-treatment variable $\mathcal{C} = \{M(1), M(0)\}$ and comparing:

$$\Pr\{Y(1) \mid M(1) = m_1, M(0) = m_0, X = x\} \quad \text{vs.} \quad \Pr\{Y(0) \mid M(1) = m_1, M(0) = m_0, X = x\}$$

- This compares $Y(1)$ vs. $Y(0)$ for the same subset of units with $\{M(1) = m_1, M(0) = m_0, X = x\}$
  - called principal stratification
  - $\{M(1), M(0)\}$ is treated as a pre-treatment covariate

Principal Stratification Average Causal Effect

- We define the principal stratification average causal effect as:

$$\tau(m_1, m_0) = E\{Y(1) - Y(0) \mid M(1) = m_1, M(0) = m_0\}$$

- For a binary M, we have four subgroups:

$$\begin{cases} \tau(1,1) &= E\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 1\}, \\ \tau(1,0) &= E\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 0\}, \\ \tau(0,1) &= E\{Y(1) - Y(0) \mid M(1) = 0, M(0) = 1\}, \\ \tau(0,0) &= E\{Y(1) - Y(0) \mid M(1) = 0, M(0) = 0\}. \end{cases}$$

Example 1: Noncompliance

Example (Noncompliance)

In randomized experiments with noncompliance, we can use M to represent the treatment received, which is affected by the treatment assignment A and affects the outcome Y.

With noncompliance, $\tau(m_1, m_0)$ represents the average causal effects for:

- $\tau(1,1)$: Always-takers
- $\tau(1,0)$: Compliers
- $\tau(0,1)$: Defiers
- $\tau(0,0)$: Never-takers

		$M(1)$	
		0	1
$M(0)$	0	Never-takers	Compliers
	1	Defiers	Always-takers

- We will return to this case during the lecture on instrumental variables (IV)

Example 2: Truncation by Death

Example (Truncation by Death)

In randomized experiments to patients with severe diseases, some patients may die before the measurement of the outcome Y, e.g., the quality of life. The post-treatment variable M in this example is the binary indicator of the survival status.

- The outcome is only well-defined if the patient survives, i.e., $M = 1$
- Hence, $\tau(1,0)$, $\tau(0,1)$ and $\tau(0,0)$ are not meaningful
- The only well-defined subgroup effect is the *survivor average causal effect* (Rubin, 2006):

$$\tau(1,1) = E\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 1\}$$

- This represents the average causal effect for those units who survive, regardless of treatment, i.e., superman

Example 3: Surrogate Endpoint

Example (Surrogate Endpoint)

In clinical trials, the outcomes of interest (e.g., 30 years of survival) require a long and costly follow-up. Practitioners instead collect data on some other variables early in the follow-up that are easy to measure. These variables are called the **surrogate endpoints**. A concrete example is from clinical trials on HIV patients, where the candidate surrogate endpoint M is the CD4 cell count (recall that CD4 cells are white blood cells that fight infection).

A good surrogate endpoint should satisfy:

- **Causal necessity**: If the treatment does not affect the surrogate, it does not affect the outcome
  $\Rightarrow \tau(1,1)$ and $\tau(0,0)$ are zero
- **Causal sufficiency**: If the treatment affects the surrogate, it must also affect the outcome
  $\Rightarrow \tau(1,0)$ and $\tau(0,1)$ are non-zero

Statistical Inference Challenges

- Even though A is randomized, we cannot identify each principal stratum from the observed data

		$M(1)$	
		0	1
$M(0)$	0	NT	CO
	1	DF	AT

Counterfactual World

		M	
		0	1
A	0	NT+CO	AT+DF
	1	NT+DF	AT+CO

Factual World

- Linear system perspective:

Number of population parameters $>$ Number of parameters identifiable from the observed data

Truncation by Death with Binary Outcome

- We use a simple setting with:
  - no X
  - binary treatment
  - binary survival status
  - binary outcome

  to illustrate the difficulty of statistical inference based on principal stratification

- In addition to Assumption $Z \perp\!\!\!\perp \{M(1), M(0), Y(1), Y(0)\}$, we also impose the monotonicity assumption:

Assumption (Monotonicity)

No defier: $M(1) \geqslant M(0)$

- There is no patient who would survive without treatment but die if treated (= no medical malpractice)

Identifying Latent Strata Proportions

		M	
		0	1
A	0	NT+CO	AT+DF
	1	NT+DF	AT+CO

We can identify the proportions of the three latent strata by:

$$\begin{aligned}
\pi(\mathsf{AT}) &= \Pr(M = 1 \mid A = 0), \\
\pi(\mathsf{NT}) &= \Pr(M = 0 \mid A = 1), \\
\pi(\mathsf{CO}) &= \Pr(M = 1 \mid A = 1) - \Pr(M = 1 \mid A = 0) \\
&= \Pr(M = 0 \mid A = 0) - \Pr(M = 0 \mid A = 1)
\end{aligned}$$

Identifying the Survivor Average Causal Effect

- Our goal is to identify the survivor average causal effect
  $\tau(\mathsf{AT}) = E\{Y(1) \mid M(1) = 1, M(0) = 1\} - E\{Y(0) \mid M(1) = 1, M(0) = 1\}$

		M	
		0	1
A	0	NT+CO	AT+DF
	1	NT+DF	AT+CO

- We can identify $E\{Y(0) \mid M(1) = 1, M(0) = 1\} = E(Y \mid A = 0, M = 1)$
  since the observed group $(A = 0, M = 1)$ consists only of AT

- How to identify $E\{Y(1) \mid \mathsf{AT}\} = E\{Y(1) \mid M(1) = 1, M(0) = 1\}$?

- The observed group $(A = 1, M = 1) = \mathsf{AT{+}CO}$ is a mixture of AT and CO:

$$E(Y \mid A = 1, M = 1) = E(Y \mid \mathsf{AT{+}CO}) = \frac{\pi(\mathsf{AT})}{\pi(\mathsf{AT}) + \pi(\mathsf{CO})} E\{Y(1) \mid \mathsf{AT}\} + \frac{\pi(\mathsf{CO})}{\pi(\mathsf{AT}) + \pi(\mathsf{CO})} E\{Y(1) \mid \mathsf{CO}\}$$

- We have two unknown parameters but only one equation

- Therefore, we cannot uniquely determine $E\{Y(1) \mid \mathsf{AT}\}$

Partial Identification

- Not identifiable $\Rightarrow$ Construct a bound on $\mathrm{E}\{Y(1)|\mathsf{AT}\}$ via linear program

$$\min / \max \Big[\mathrm{E}\{Y(1)|\mathsf{AT}\}\Big]$$

subject to
$$E(Y \mid A=1, M=1) = \frac{\pi(\mathsf{AT})}{\pi(\mathsf{AT})+\pi(\mathsf{CO})} E\{Y(1) \mid \mathsf{AT}\} + \frac{\pi(\mathsf{CO})}{\pi(\mathsf{AT})+\pi(\mathsf{CO})} E\{Y(1) \mid \mathsf{CO}\}$$
$$E\{Y(1) \mid \mathsf{AT}\},\ E\{Y(1) \mid \mathsf{CO}\} \in [0,1]$$

- Solution:

$$\mathrm{E}\{Y(1)|\mathsf{AT}\} \in \left[\frac{\{\pi(\mathsf{AT})+\pi(\mathsf{CO})\}E(Y \mid A=1, M=1) - \pi(\mathsf{CO})}{\pi(\mathsf{AT})}, \frac{\{\pi(\mathsf{AT})+\pi(\mathsf{CO})\}E(Y \mid A=1, M=1)}{\pi(\mathsf{AT})} \right]$$

- A 95% uncertainty interval for this bound can also be constructed for inference:
  - Imbens and Manski (2004): Simple, but required conditions are too strong to be satisfied
  - Chernozhukov et al. (2013): Applicable to most bounds, but hard to implement and verify
  - Levis et al. (2025): Uses semiparametric efficiency theory, but requires a closed-form solution to the LP

Application: ARDS Network Study

- We use the data from Yang and Small (2016) based on the Acute Respiratory Distress Syndrome (ARDS) Network study
- The study involved 861 patients with lung injury and acute respiratory distress syndrome
- Patients were randomized to receive mechanical ventilation with:
  - Lower tidal volumes, or
  - Traditional tidal volumes
- The outcome of interest is a binary indicator for whether patients could breathe without assistance by day 28

Observed Data Summary

The observed data, summarized in Table 1, includes treatment assignment (A), post-treatment variable (M), and outcome (Y), with * indicating the outcomes for deceased patients

Table: Data Truncated by Death with * Indicating the Outcomes for Dead Patients

Treatment $A = 1$			
	$Y = 1$	$Y = 0$	total
$M = 1$	54	268	322
$M = 0$	*	*	109

Control $A = 0$			
	$Y = 1$	$Y = 0$	total
$M = 1$	59	218	277
$M = 0$	*	*	152

Estimating Causal Effects

- The estimated latent strata proportions are:

$$\hat{\pi}(\mathsf{AT}) = \frac{277}{277+152} = 0.646 \qquad \hat{\pi}(\mathsf{NT}) = \frac{109}{109+322} = 0.253 \qquad \hat{\pi}(\mathsf{CO}) = 1 - 0.646 - 0.253 = 0.101$$

- The sample means for the surviving patients are:

$$\hat{E}(Y \mid A=1, M=1) = \frac{54}{302} = 0.168 \qquad \hat{E}(Y \mid A=0, M=1) = \frac{59}{277} = 0.213$$

- The bounds for $E\{Y(1) \mid M(1)=1, M(0)=1\}$ are

$$\left[\frac{(0.646+0.101) \times 0.168 - 0.101}{0.101} \;,\; \frac{(0.646+0.101) \times 0.168}{0.101} \right] = [0.037, 0.194]$$

  leading to bounds on $\tau(\mathsf{AT})$:

$$[0.037 - 0.213, 0.194 - 0.213] = [-0.176, -0.019]$$

- Incorporating bootstrap uncertainty, the confidence interval for $\tau(1,1)$ is $[-0.267, 0.039]$, which covers zero

Point Identification of Principal Stratification Average Causal Effect

- We assume

Assumption

- *Complete Randomization:* $A \perp\!\!\!\perp \{M(1), M(0), Y(1), Y(0), X\}$
- *Strong Monotonicity:* $M(0) = 0$
- *Principal Ignorability:* $E\{Y(0) \mid M(1) = 1, X\} = E\{Y(0) \mid M(1) = 0, X\}$

- Strong Monotonicity:
  - Strong monotonicity implies monotonicity
  - There are two subgroups: CO $= \{M(1) = 1, M(0) = 0\}$ and NT $= \{M(1) = 0, M(0) = 0\}$

- Principal Ignorability
  - It implies $E\{Y(0) \mid M(1), X\} = E\{Y(0) \mid X\}$ or equivalently $E\{Y(0)M(1) \mid X\} = E\{Y(0) \mid X\}E\{M(1) \mid X\}$ that is, $Y(0)$ and $M(1)$ are mean independent (or uncorrelated) given X.

Identification

Theorem (Identification of Principal Stratification Causal Effects)

Under these three assumptions, the following results hold:

- $\pi(X) = \Pr\{M(1) = 1 \mid X\}$ *and* $\pi = \Pr\{M(1) = 1\}$ *are identified by:*

$$\pi(X) = \Pr(M = 1 \mid A = 1, X), \quad \pi = \Pr(M = 1 \mid A = 1).$$

- *The principal stratification average causal effects can be identified as:*

$$\tau(1,0) = \tau(\mathsf{CO}) = E(Y \mid A = 1, M = 1) - \frac{E\{\pi(X)Y \mid A = 0\}}{\pi},$$

$$\tau(0,0) = \tau(\mathsf{NT}) = E(Y \mid A = 1, M = 0) - \frac{E[\{1 - \pi(X)\}Y \mid A = 0]}{1 - \pi}.$$

- The conditional probability $\pi(X) = \Pr\{M(1) = 1 \mid X\}$ is known as the *principal score*.

Proof for Identifying $\tau(1,0)$

- Note that

$$\tau(1,0) = \mathrm{E}\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 0\} = \mathrm{E}\{Y(1) - Y(0) \mid M(1) = 1\}$$

- We have $\mathrm{E}[Y(1) \mid M(1) = 1] = \mathrm{E}[Y(1) \mid A = 1, M(1) = 1] = \mathrm{E}[Y \mid A = 1, M = 1]$
- So it suffices to obtain an identifying formula for $\mathrm{E}[Y(0) \mid M(1) = 1]$

$$\begin{aligned}
\mathrm{E}\{Y(0) \mid M(1) = 1\} &= \mathrm{E}\{M(1)Y(0)\}/\pi \\
&= \mathrm{E}[\mathrm{E}\{M(1)Y(0) \mid X\}]/\pi \\
&= \mathrm{E}[\mathrm{E}\{M(1) \mid X\}\mathrm{E}\{Y(0) \mid X\}]/\pi && \text{Principal Ignorability} \\
&= \mathrm{E}[\pi(X)\mathrm{E}\{Y(0) \mid X\}]/\pi \\
&= \mathrm{E}\{\pi(X)Y(0)\}/\pi \\
&= \mathrm{E}\{\pi(X)Y(0) \mid A = 0\}/\pi && \text{Complete Randomization} \\
&= \mathrm{E}\{\pi(X)Y \mid A = 0\}/\pi
\end{aligned}$$

Estimating $\tau(1,0)$ and $\tau(0,0)$

- To estimate the causal effects $\tau(1,0)$ and $\tau(0,0)$, follow these steps:

1. Fit a logistic regression of M on X using only data from the treated group to obtain $\hat{\pi}(X_i)$
2. Estimate π as:

$$\hat{\pi} = \frac{\sum A_i M_i}{\sum A_i}$$

3. Use moment estimators for $\tau(1,0)$ and $\tau(0,0)$:

$$\hat{\tau}(1,0) = \frac{\sum A_i M_i Y_i}{\sum A_i M_i} - \frac{\sum (1-A_i)\hat{\pi}(X_i) Y_i}{\hat{\pi} \sum (1-A_i)}$$

$$\hat{\tau}(0,0) = \frac{\sum A_i (1-M_i) Y_i}{\sum A_i (1-M_i)} - \frac{\sum (1-A_i)\{1-\hat{\pi}(X_i)\} Y_i}{(1-\hat{\pi}) \sum (1-A_i)}$$

4. Use bootstrap to approximate the variances of $\hat{\tau}(1,0)$ and $\hat{\tau}(0,0)$

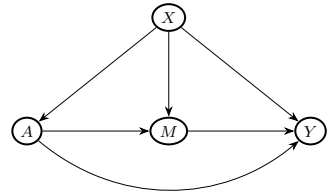
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Goal of Mediation Analysis

- We define the principal stratification average causal effect as:

$$\tau(m_1, m_0) = E\{Y(1) - Y(0) \mid M(1) = m_1, M(0) = m_0\}$$



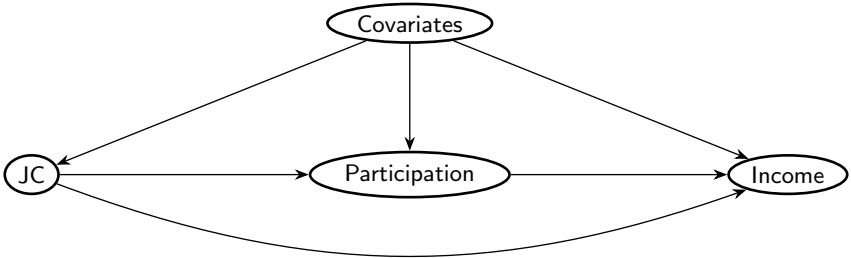
- For a binary M, we have four subgroups:

$$\begin{cases} \tau(1,1) &= E\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 1\} \\ \tau(1,0) &= E\{Y(1) - Y(0) \mid M(1) = 1, M(0) = 0\} \\ \tau(0,1) &= E\{Y(1) - Y(0) \mid M(1) = 0, M(0) = 1\} \\ \tau(0,0) &= E\{Y(1) - Y(0) \mid M(1) = 0, M(0) = 0\} \end{cases}$$

  - $\tau(1,1)$ and $\tau(0,0)$ reflect direct effects
  - $\tau(1,0)$ and $\tau(0,1)$ combine direct and indirect effects
  - Principal stratification cannot isolate indirect effects, as it does not assume that M can be intervened

- Mediation analysis formally quantifies:
  - The effect of A on Y through M
  - The effect of A on Y via other pathways

Example: Job Corps Program



- Data: `JC` dataset from the `causalweight` package in R
- Context: Evaluation of the Job Corps program
- Treatment: Recommendation to participate in Job Corps (year 1)
- Mediator: Participation in vocational training (year 2)
- Outcomes: Weekly income and working hours (year 4)
- Objective: Assess the indirect effects (through vocational training participation) and the direct effects of the intervention

Potential Outcomes and Interventions

- Binary treatment $A \in \{0, 1\}$
- $\{Y_i, M_i, A_i, X_i\}$ for N units from a superpopulation
- Define potential outcomes for intervention on A:

$$\{M(a), Y(a) \,|\, a = 0, 1\}$$

- Define potential outcomes for intervention on both A and M:

$$\{Y(a, m) \,|\, a = 0, 1,\ m \in \mathcal{M}\}$$

- Define nested potential outcomes:

$$\{Y(z, M_{a'}) \,|\, a = 0, 1;\ a' = 0, 1\} \quad \text{where } M(a') = M_{a'}$$

- Example with binary treatment:

$$\{Y(1, M_1), Y(1, M_0), Y(0, M_1), Y(0, M_0)\}$$

Definition of Effects

- Composition assumption: $Y(a, M_a) = Y(a)$ for $a = 0, 1$
  - Cannot be tested or proven
  - Can be taken as definitional: define $Y(a) := Y(a, M_a)$

- $Y(a, M_a)$ for $a = 0, 1$: treatment set to a, mediator set to its natural value under $A = a$
  - Single-world counterfactuals, can be represented in SWIGs

- $Y(1, M_0)$: treatment set to 1, mediator set to M_0; $Y(0, M_1)$: treatment set to 0, mediator set to M_1
  - Cross-world counterfactuals, cannot be represented in SWIGs

Natural Direct/Indirect Effects

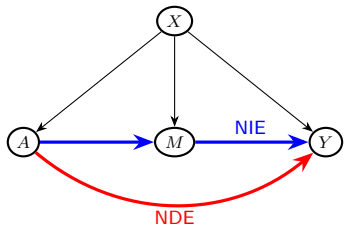
Definition (Total, Direct, and Indirect Effects; Pearl (2001))

$$\text{ATE} = E\{Y(1) - Y(0)\} \quad \text{(Total Effect)}$$

$$\text{NDE} = E\{Y(1, M_0) - Y(0, M_0)\} \quad \text{(Natural Direct Effect)}$$

$$\text{NIE} = E\{Y(1, M_1) - Y(1, M_0)\} \quad \text{(Natural Indirect Effect)}$$

- ATE: average causal effect of A on Y
- NDE: effect of A on Y with M held at M_0
- NIE: effect of changing M under $A = 1$
- Effect decomposition: ATE = NDE + NIE
- We call $\mathrm{E}\{Y(1, M_0)\}$ the mediation functional



Sequential Ignorability

- Summarizing Pearl (2001); Imai et al. (2010)

Assumption (Sequential Ignorability)

Treatment Ignorability :	$\{M(a), Y(a', m)\} \perp\!\!\!\perp A \,\big|\, X$,	$\forall(a, a', m)$
Cross-world Independence :	$Y(a', m) \perp\!\!\!\perp M(a) \,\big|\, (A = a, X)$,	$\forall(a, a', m)$

- Cross-world independence cannot be verified even in randomized experiments (not a single-world assumption)

- However, under the NPSEM-IEs, cross-world independence is satisfied:

$$\begin{aligned} Y(a', m) &= g_Y(a', m, X, \epsilon_Y) \\ M(a) &= g_M(a, X, \epsilon_M) \\ A &= g_A(X, \epsilon_A) \\ X &= g_X(\epsilon_X) \end{aligned} \qquad \text{where} \quad \epsilon_Y \perp\!\!\!\perp \epsilon_M \perp\!\!\!\perp \epsilon_A \perp\!\!\!\perp \epsilon_X$$

Identification

Assumption (Sequential Ignorability)

Treatment Ignorability : $\{M(a), Y(a', m)\} \perp\!\!\!\perp A \,|\, X \ , \qquad \forall (a, a', m)$

Cross-world Independence : $Y(a', m) \perp\!\!\!\perp M(a) \,|\, (A = a, X) \ , \qquad \forall (a, a', m)$

Assumption (Positivity in the Mediation Setting)

$$\Pr(A = a|X) > 0 \ , \quad \forall (a, X) \ , \qquad \Pr(M = m|A = a, X) > 0 \ , \quad \forall (m, a, X)$$

Theorem (Pearl (2001))

Under the sequential ignorability and the positivity assumption:

$$E\{Y(a, M_{a'}) \mid X = x\} = \sum_m E(Y \mid A = a, M = m, X = x) P(M = m \mid A = a', X = x)$$

and therefore,

$$\begin{aligned} E\{Y(a, M_{a'})\} &= \sum_x E\{Y(a, M_{a'}) \mid X = x\} P(X = x) \\ &= \sum_{x,m} E(Y \mid A = a, M = m, X = x) P(M = m \mid A = a', X = x) P(X = x). \end{aligned}$$

Identification

Theorem (Pearl (2001))

Under the sequential ignorability and the positivity assumption:

$$E\{Y(a, M_{a'}) \mid X = x\} = \sum_m E(Y \mid A = a, M = m, X = x) P(M = m \mid A = a', X = x)$$

and therefore,

$$\begin{aligned} E\{Y(a, M_{a'})\} &= \sum_x E\{Y(a, M_{a'}) \mid X = x\} P(X = x) \\ &= \sum_{x,m} E(Y \mid A = a, M = m, X = x) P(M = m \mid A = a', X = x) P(X = x) \end{aligned} \tag{1}$$

- Identification requires:
  - Outcome regression: $E(Y \mid Z, M, X)$:
  - Mediator model: $P(M \mid Z, X)$
- Both models evaluated at different treatment levels
- Cross-world nature still encoded in the integral

Identification

$$
\begin{aligned}
E\{Y(a, M_{a'}) \mid X = x\} &= \sum_m E\{Y(a, M_{a'}) \mid M_{a'} = m, X = x\} P(M_{z'} = m \mid X = x) \\
&= \sum_m E\{Y(a, m) \mid M_{a'} = m, X = x\} P(M_{a'} = m \mid X = x) \\
&= \sum_m E\{Y(a, m) \mid X = x\} P(M = m \mid A = a', X = x) \qquad \text{Seq. Ign.} \\
&= \sum_m E(Y \mid A = a, M = m, X = x) P(M = m \mid A = a', X = x)
\end{aligned}
$$

Mediation Formula

$$E\{Y(1, M_1) \mid X = x\} = \sum_m E(Y \mid A = 1, M = m, X = x) P(M = m \mid A = 1, X = x)$$
$$= E(Y \mid A = 1, X = x)$$

$$E\{Y(0, M_0) \mid X = x\} = \sum_m E(Y \mid A = 0, M = m, X = x) P(M = m \mid A = 0, X = x)$$
$$= E(Y \mid A = 0, X = x)$$

$$E\{Y(1, M_0) \mid X = x\} = \sum_m E(Y \mid A = 1, M = m, X = x) P(M = m \mid A = 0, X = x)$$

This mixes the conditional expectation under $A = 1$ with the distribution of M under $A = 0$

Identification of Conditional Effects

Corollary (Conditional Natural Effects)

The conditional and unconditional NDE and NIE are identified by

$$\textit{NDE}(x) = \sum_m \Big\{ E(Y \mid A = 1, M = m, X = x) - E(Y \mid A = 0, M = m, X = x) \Big\} \Pr(M = m \mid Z = 0, X = x)$$

$$\textit{NIE}(x) = \sum_m E(Y \mid A = 1, M = m, X = x) \Big\{ \Pr(M = m \mid A = 1, X = x) - \Pr(M = m \mid A = 0, X = x) \Big\}$$

and

$$\textit{NDE} = \sum_x \textit{NDE}(x) P(X = x)$$

$$\textit{NIE} = \sum_x \textit{NIE}(x) P(X = x).$$

Special Case: Binary Mediator

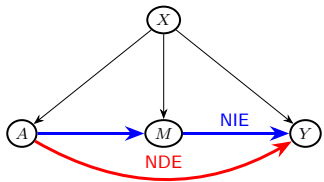
Corollary (Binary Mediator)

Assume $M \in \{0, 1\}$. *Then* NIE$(x) = \tau_{A\to M}(x) \times \tau_{M\to Y}(1, x)$ *and* NIE $= E\{$NIE$(X)\}$ *where*

$$\tau_{A\to M}(x) = P(M = 1 \mid A = 1, X = x) - P(M = 1 \mid A = 0, X = x),$$

$$\tau_{A\to Y}(a, x) = E(Y \mid A = a, M = 1, X = x) - E(Y \mid A = a, M = 0, X = x)$$

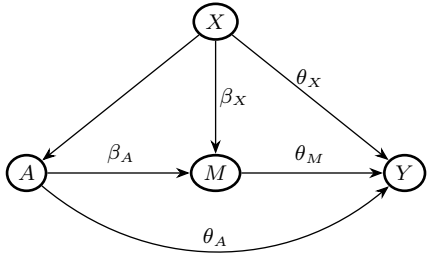
- Randomization of A given X $\Rightarrow \tau_{A\to M}(x)$: conditional average causal effect of A on M
- Randomization of M given X, A $\Rightarrow \tau_{A\to Y}(a, x)$: conditional average causal effect of M on Y
- NIE(x) equals their product $\Rightarrow$ intuitively capturing the indirect pathway from A to Y through M



The Baron-Kenny Method

Assumption (Linear models for mediator and outcome)

$$E(M \mid A, X) = \beta_0 + \beta_A A + \beta_X^\top X$$
$$E(Y \mid A, M, X) = \theta_0 + \theta_A A + \theta_M M + \theta_X^\top X$$



- Then, NDE $= \theta_A$, NIE $= \theta_M \beta_A$
- Use OLS to estimate linear coefficients: $\widehat{\text{NDE}} = \hat{\theta}_A$ and $\widehat{\text{NIE}} = \hat{\theta}_M \hat{\beta}_A$
- Baron-Kenny method (Baron and Kenny, 1986; Judd and Kenny, 1981)
- $\widehat{\text{NIE}} = \hat{\theta}_M \hat{\beta}_A$ is a product estimator
- Sobel (1982) used the delta method:

$$\text{Var}(\widehat{\text{NIE}}) \approx \text{Var}(\hat{\theta}_2)\beta_1^2 + \theta_2^2 \text{Var}(\hat{\beta}_1) \quad \Rightarrow \quad \widehat{\text{Var}}(\widehat{\text{NIE}}) = \widehat{\text{Var}}(\hat{\theta}_2)\hat{\beta}_1^2 + \hat{\theta}_2^2 \widehat{\text{Var}}(\hat{\beta}_1)$$

EIF-based Estimator

- Tchetgen Tchetgen and Shpitser (2012) shows that the EIF of the mediation functional $\psi := \mathrm{E}\{Y(1, M_0)\}$ in the nonparametric model is given by

$$\begin{aligned}&\mathsf{EIF}(O;\psi)\\ &= \frac{A}{e(X)}\frac{f_{M|AX}(M|A=0,X)}{f_{M|AX}(M|A=1,X)}\{Y - \mathrm{E}(Y|A=1,M,X)\} + \frac{1-A}{1-e(X)}\{\mathrm{E}(Y|A=1,M,X) - \mu(X)\} + \mu(X) - \psi\end{aligned}$$

where

$$e(X) = \Pr(A=1|X)\ , \qquad \mu(X) = \int \mathrm{E}(Y|A=1,M=m,X) f_{M|A,X}(m|A=0,X)\, dm$$

- This approach involves three nuisance functions:

$$\eta_1 := \mathrm{E}(Y|A=1,M,X) \qquad \eta_2 := f_{M|A,X}(M|A,X) \qquad \eta_3 := e(X)$$

- **Multiple robustness**: as long as two out of three nuisance components (e.g., η_1 and η_2) are correctly specified, the EIF-based estimator $\widehat{\psi}_{\mathsf{EIF}}$ is consistent for ψ

Example

- Job Search Intervention Study (JOBS II) in `mediation::jobs`
- A randomized experiment that investigates the efficacy of a job training intervention on unemployed workers
- 1,801 unemployed workers received a pre-screening questionnaire and were randomly assigned to trt/ctrl groups
- Trt group: participated in job-skills workshops in order to learn job-search skills and coping strategies
- Ctrl group: received a booklet describing job-search tips
- Mediator: level of job-search self-efficacy (continuous)
- Outcome:
  - measure of depressive symptoms (continuous)
  - employment (binary)

- We will use `mediation` R package (Tingley et al., 2014)

Example

```
library(mediation)
data(jobs)

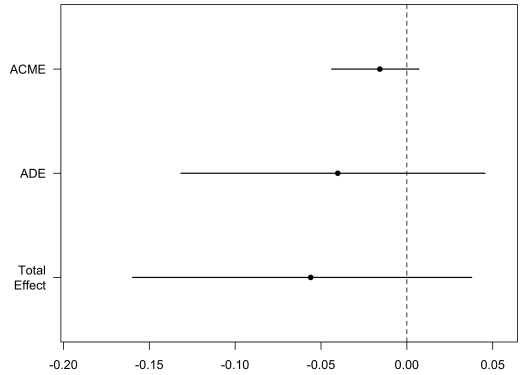
# Linear mediator model
med.noint <- lm(job_seek ~ treat + econ_hard + sex + age, data=jobs)

# Linear outcome model
OR.cont.noint <- lm(depress2 ~ treat + job_seek + econ_hard + sex + age, data=jobs)

# NDE/NIE under linear models with conti Y
Result.cont.noint <- mediate(med.noint, OR.cont.noint, boot=TRUE, sims=500, treat="treat", mediator="job_seek")
summary(Result.cont.noint)
plot(Result.cont.noint)
```

Example

	Estimate	95% CI Lower	95% CI Upper	p-value
ACME	-0.0157	-0.0382	0.01	0.17
ADE	-0.0403	-0.1282	0.05	0.42
Total Effect	-0.0560	-0.1383	0.04	0.24
Prop. Mediated	0.2811	-1.7262	2.50	0.32



Example

```
# Interaction mediator model
med.int <- lm(job_seek ~ (treat + econ_hard + sex + age)^2, data=jobs)

# Interaction outcome model
OR.cont.int <- lm(depress2 ~ (treat + job_seek + econ_hard + sex + age)^2, data=jobs)

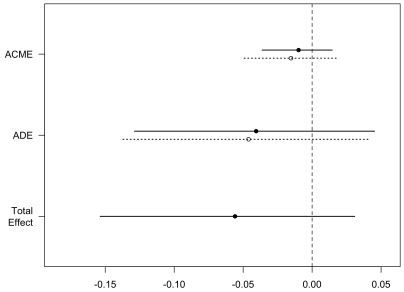
# NDE/NIE under interaction models with conti Y
Result.cont.int <- mediate(med.int, OR.cont.int, boot=TRUE, sims=500, treat="treat", mediator="job_seek")


# GLM
OR.binary.noint <- glm(work1 ~ treat + job_seek + econ_hard + sex + age, family="binomial", data=jobs)

# NDE/NIE under linear models with binary Y
Result.binary.noint <- mediate(med.noint, OR.binary.noint, boot=TRUE, sims=500, treat="treat", mediator="job_seek")
```

Example

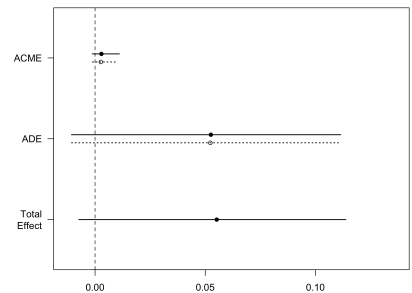
	Estimate	95% CI Lower	95% CI Upper	p-value
ACME (control)	-0.01531	-0.04925	0.02	0.26
ACME (treated)	-0.00979	-0.03620	0.01	0.32
ADE (control)	-0.04609	-0.13714	0.04	0.33
ADE (treated)	-0.04057	-0.12875	0.05	0.36
Total Effect	-0.05588	-0.15366	0.03	0.26
Prop. Mediated (control)	0.27396	-2.09095	2.51	0.37
Prop. Mediated (treated)	0.17518	-1.61630	1.60	0.42
ACME (average)	-0.01255	-0.04148	0.02	0.28
ADE (average)	-0.04333	-0.13164	0.04	0.35
Prop. Mediated (average)	0.22457	-1.93172	2.19	0.39



- Bold line: $\mathrm{E}\{Y(1, M_1) - Y(1, M_0)\}$ and $\mathrm{E}\{Y(1, M_1) - Y(0, M_1)\}$
- Dashed line: $\mathrm{E}\{Y(0, M_1) - Y(0, M_0)\}$ and $\mathrm{E}\{Y(1, M_0) - Y(0, M_0)\}$

Example

	Estimate	95% CI Lower	95% CI Upper	p-value	
ACME (control)	0.00265	-0.00116	0.01	0.240	
ACME (treated)	0.00289	-0.00126	0.01	0.240	
ADE (control)	0.05232	-0.01064	0.11	0.100	.
ADE (treated)	0.05256	-0.01068	0.11	0.100	.
Total Effect	0.05521	-0.00736	0.11	0.092	.
Prop. Mediated (control)	0.04795	-0.09349	0.40	0.284	
Prop. Mediated (treated)	0.05228	-0.09431	0.41	0.284	
ACME (average)	0.00277	-0.00122	0.01	0.240	
ADE (average)	0.05244	-0.01066	0.11	0.100	.
Prop. Mediated (average)	0.05011	-0.09390	0.40	0.284	



Outline

1. Overview
2. Principal Stratification
3. Pearl's Natural Direct/Indirect Effects
4. Robins and Greenland's Controlled Direct Effect
5. Treatment Decomposition Approach
6. Summary
7. Assignment

Definition of the Controlled Direct Effect

Definition (Controlled Direct Effect; Robins and Greenland (1992))

For fixed $m \in \mathcal{M}$:

$$\text{CDE}(m) = E\{Y(1, m) - Y(0, m)\}$$

- CDE captures the causal effect of Z on Y when mediator M is *fixed* at m
- Cannot define indirect effects in this framework
- $E\{Y(a, 1) - Y(a, 0)\}$ only captures effect of M on Y holding $A = a$, so this is not a valid indirect effect
- This quantity is instead referred to as the controlled mediator effect (Zheng and Zhou, 2014)

Identification of CDE(m)

Assumption (Sequential Ignorability)

Treatment Ignorability :	$\{M(a), Y(a,m)\} \perp\!\!\!\perp A \,\big|\, X$,	$\forall (a,m)$
No Unmeasured Confounding in Y-M :	$Y(a,m) \perp\!\!\!\perp M(a) \,\big|\, (A=a, X)$,	$\forall (a,m)$

- Binary (A, M): treat (A, M) as a joint treatment with four levels: $(0,0), (0,1), (1,0), (1,1)$
- Goal: identify $\mu_{am} = E\{Y(a,m)\}$

Identification Theorem

Assumption (Positivity in the Mediation Setting)

$$\Pr(A = a|X) > 0 \ , \quad \forall(a, X) \ , \qquad \Pr(M = m|A = a, X) > 0 \ , \quad \forall(m, a, X)$$

Theorem (Identification of μ_{am})

Under the sequential ignorability assumption and the positivity assumption:

$$\mu_{am} = E\{\mu_{am}(X)\} = E\left\{\frac{\mathbb{1}(A = a, M = m)Y}{e_{am}(X)}\right\}$$

where:

$$\mu_{am}(X) = E(Y \mid A = a, M = m, X)$$

$$e_{am}(X) = \Pr(A = a, M = m \mid X) = \Pr(A = a \mid X) \cdot \Pr(M = m \mid A = a, X)$$

EIF-based Estimator of μ_{am}

- The EIF of μ_{am} in the nonparametric model is

$$\mathsf{EIF}(O;\mu_{am}) = \frac{\mathbb{1}(A=a,M=m)\{Y-\mu_{am}(X)\}}{e_{am}(X)} + \{mu_{am}(X)-\mu_{am}\}$$

- Given working models $\mu_{am}(X,\beta)$ and $e_{am}(X,\alpha)$:

$$\widehat{\mu}_{am}^{\mathsf{dr}} = \mathbb{P}_N\left[\frac{\mathbb{1}(A=a,M=m)\{Y-\mu_{am}(X,\beta)\}}{e_{am}(X,\alpha)} + \mu_{am}(X,\beta)\right]$$

- Consistent if either outcome model or propensity model is correctly specified

- In turn, one can construct an estimator for CDE:

$$\widehat{\mathsf{CDE}}(m) = \widehat{\mu}_{1m}^{\mathsf{dr}} - \widehat{\mu}_{0m}^{\mathsf{dr}}$$

  where the standard error can be obtained from nonparametric bootstrap

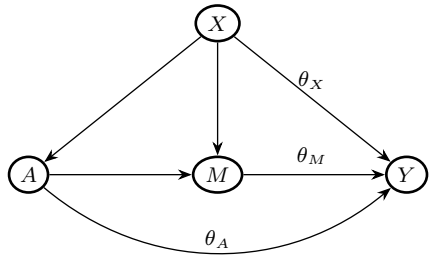
- One can also consider IPW and OR estimators in a similar manner

- Re-visit lecture notes for IPW/OR/DR estimators

Linear Outcome Model

Assumption (Linear models for mediator and outcome)

$$E(Y \mid A, M, X) = \theta_0 + \theta_A A + \theta_M M + \theta_X^\top X$$



- Then, CDE$(m) = \theta_A$, which equals to the NDE

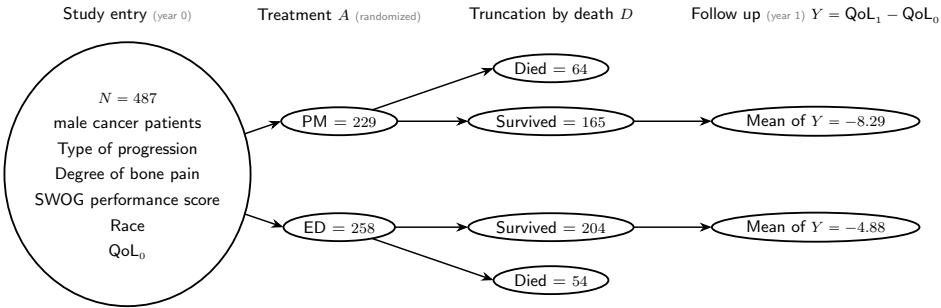
- The same result holds for the partially linear model

$$\mathrm{E}(Y \,|\, A, M, X) = \theta_A A + f(M, X)$$

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Motivating Example: Southwest Oncology Group (SWOG) Trial



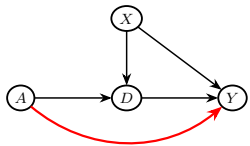
Treatment	Pain reliever (palliation; may have side effects)	Chemotherapeutic agent (reduce proliferation of cancer cells)
P+M	**P**rednisone	**M**itoxantrone
E+D	**E**stramustine	**D**ocetaxel

Candidate Causal Effects

- Candidate 1: **Controlled direct effect**

$$\mathrm{E}\{Y(a = \mathsf{P{+}M}, d = 0) - Y(a = \mathsf{E{+}D}, d = 0)\}$$

  - causal effect on Y in a hypothetical world where death can be prevented



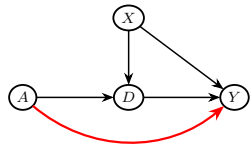
- Limitations:
  - relies on setting $d = 0$ (i.e., which everyone survives the cancer); not available, not meaningful

Candidate Causal Effects

- Candidate 2: **Pure (natural) direct effects**

$$\mathrm{E}\{Y(a = \mathsf{P{+}M}, d = D(a = \mathsf{P{+}M})) - Y(a = \mathsf{E{+}D}, d = D(a = \mathsf{P{+}M}))\}$$

  - causal effect on Y when D is allowed to take the level it would naturally attain under $a = \mathsf{P} + \mathsf{M}$



- Limitations:

  defined w.r.t interventions that cannot be implemented

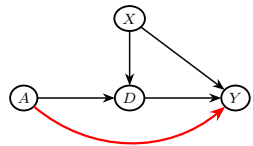
  relies on cross-world independence assumptions (untestable even in randomized experiments)

Candidate Causal Effects

- Candidate 3: **Principal stratum effects**

$$\mathrm{E}\{Y(a = \mathsf{P{+}M}) - Y(a = \mathsf{E{+}D}) \,|\, D(a = \mathsf{P{+}M}) = D(a = \mathsf{E{+}D}) = 0\}$$

  - causal effect on Y among always survivors



- Limitations:
  - always survivors cannot be observed since we never see survival under both treatments for one person
  - always survivors may not actually exist
  - even if always survivors exist, they may be an unusual subgroup, making the estimand less meaningful

- Alternative: **Conditional Separable Effects** (Stensrud et al., 2022)
  based on the treatment decomposition (Robins and Richardson, 2011)

A New Future 4-arm Trial

- In the SWOG trial, treatment is a combination of a chemotherapeutic agent (A_D) and a pain reliever (A_Y):

A_Y	Pain reliever	Prednisone, Estramustine
A_D	Chemotherapeutic agent	Mitoxantrone, Docetaxel

A New Future 4-arm Trial

- In the SWOG trial, treatment is a combination of a chemotherapeutic agent (A_D) and a pain reliever (A_Y):

A_Y	Pain reliever	Prednisone, Estramustine
A_D	Chemotherapeutic agent	Mitoxantrone, Docetaxel

- Consider a new future 4-arm trial where a patient is randomly assigned to $A_D \in \{M, D\}$ and $A_Y \in \{P, E\}$

(A_Y, A_D)	P+M	E+D	E+M	P+D
Previous 2-arm trial (SWOG)	available	available	not available	not available
Future 4-arm trial	available	available	available	available

A New Future 4-arm Trial

- In the SWOG trial, treatment is a combination of a chemotherapeutic agent (A_D) and a pain reliever (A_Y):

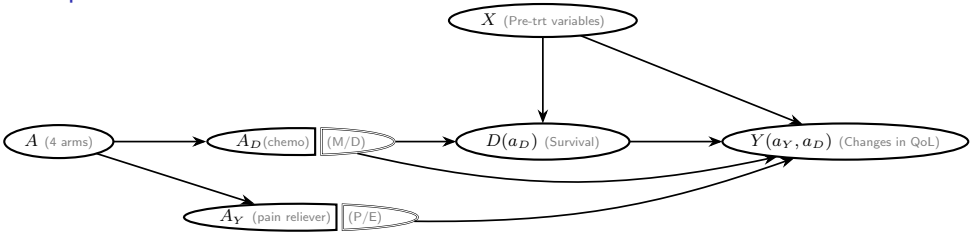
A_Y	Pain reliever	Prednisone, Estramustine
A_D	Chemotherapeutic agent	Mitoxantrone, Docetaxel

- Consider a new future 4-arm trial where a patient is randomly assigned to $A_D \in \{M, D\}$ and $A_Y \in \{P, E\}$

(A_Y, A_D)	P+M	E+D	E+M	P+D
Previous 2-arm trial (SWOG)	available	available	not available	not available
Future 4-arm trial	available	available	available	available

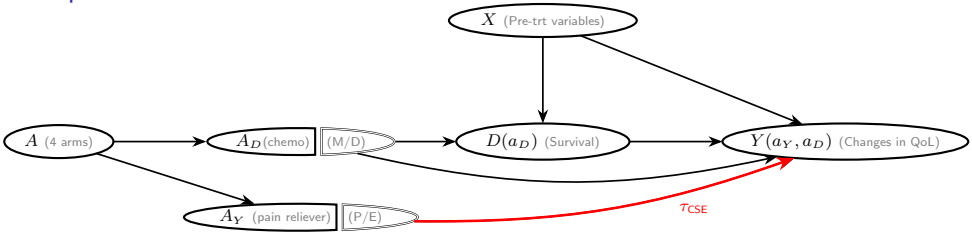
- $Y(a_Y, a_D), D(a_Y, a_D) =$ potential outcome, potential survival under $(A_Y, A_D) = (a_Y, a_D)$ in the future 4-arm trial

Conditional Separable Effects



- (A_Y *partial isolation*): there is no causal path from A_Y to D

Conditional Separable Effects



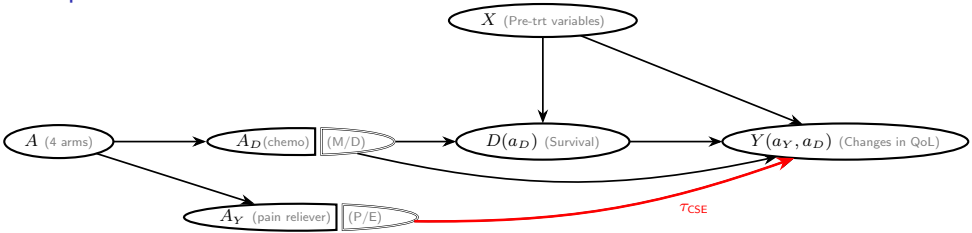
- (A_Y *partial isolation*): there is no causal path from A_Y to D

- Conditional Separable Effects:

$$\tau^*_{\mathsf{CSE}} = \mathrm{E}\big\{Y(a_Y = \mathsf{Pred}, a_D = \mathsf{Doce}) - Y(a_Y = \mathsf{Estra}, a_D = \mathsf{Doce}) \,\big|\, D(a_D = \mathsf{Doce}) = 0\big\}$$

  - Assuming all patients receive docetaxel, the causal effect of prednisone v. estramustine on QoL among survivors

Conditional Separable Effects



- (A_Y *partial isolation*): there is no causal path from A_Y to D

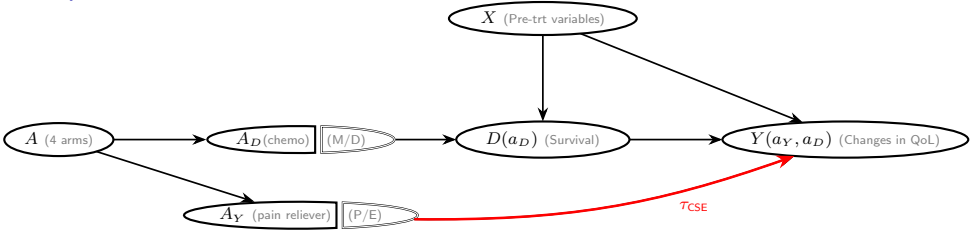
- Conditional Separable Effects:

$$\tau^*_{\mathsf{CSE}} = \mathrm{E}\big\{Y(a_Y = \mathsf{Pred}, a_D = \mathsf{Doce}) - Y(a_Y = \mathsf{Estra}, a_D = \mathsf{Doce}) \,\big|\, D(a_D = \mathsf{Doce}) = 0\big\}$$

  - Assuming all patients receive docetaxel, the causal effect of prednisone v. estramustine on QoL among survivors

- Under (A_Y partial isolation), τ^*_{CSE} captures a **direct** effect of A on Y among survivors

Conditional Separable Effects



- (A_Y *partial isolation*): there is no causal path from A_Y to D
  (A_D *partial isolation*): the only causal path from A_D to Y are directed paths intersected by D

- Conditional Separable Effects:

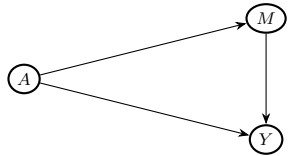
$$\tau^*_{\mathsf{CSE}} = \mathrm{E}\big\{Y(a_Y = \mathsf{Pred}, a_D = \mathsf{Doce}) - Y(a_Y = \mathsf{Estra}, a_D = \mathsf{Doce}) \,\big|\, D(a_D = \mathsf{Doce}) = 0\big\}$$

  - Assuming all patients receive docetaxel, the causal effect of prednisone v. estramustine on QoL among survivors

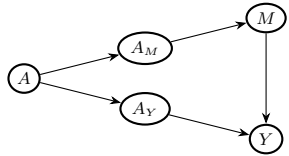
- Under (A_Y partial isolation), τ^*_{CSE} captures a **direct** effect of A on Y among survivors
  Under (A_Y and A_D partial isolation), τ^*_{CSE} captures **all** effects of A on Y among survivors

Robins and Richardson (2011)'s Original Example

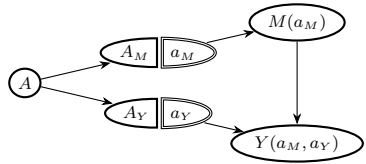
- A = Smoking status
- M = Blood pressure
- Y = Heart attack



- Suppose there are additional nodes:
  - A_M = Nicotine exposure
  - A_Y = Other chemical components of cigarettes



- These nodes are associated with well-defined interventions, s.t.,
  - the effect of A on M is via A_M
  - the effect of A on Y is via A_Y



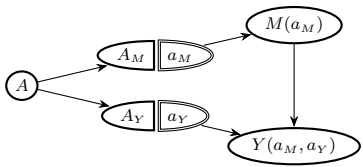
Interventional Direct/indirect Effects

- We can define **interventional direct/indirect effects**

$$\tau_{\text{IIE}} = \mathrm{E}\{Y(a_M = 1, a_Y = 1) - Y(a_M = 0, a_Y = 1)\}$$

$$\tau_{\text{IDE}} = \mathrm{E}\{Y(a_M = 0, a_Y = 1) - Y(a_M = 0, a_Y = 0)\}$$

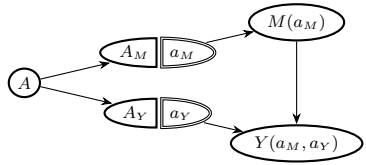
  - Effect of smoking regular cigarettes vs. nicotine-free cigarettes
  - Effect of smoking nicotine-free cigarettes vs. not smoking



- These are well-defined causal effects

Identification of Interventional Direct/indirect Effects

- Consider three experiments



(i) The original trial in which A was randomized
so the data in arm $A = a$ corresponds to $M(a)$ and $Y(a)$

(ii) A 4-arm trial where $(A_M, A_Y) \in \{0,1\}^{\otimes 2}$ are randomized
so the data in arm $(A_M, A_Y) = (a_M, a_Y)$ correspond to $M(a_M)$ and $Y(a_M, a_Y)$

(iii) A 2-arm trial from (ii) by restricting to the 2-arms with $A_M = A_Y$
so the data in arm $A = A_M = A_Y = a$ correspond to $M(a)$ and $Y(a, a)$

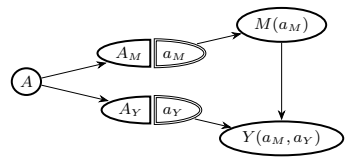
- Note that (i) and (iii) are conceptually the same
- Note that τ_{IDE} and τ_{IIE} are defined within (ii)
- When is the distribution of (ii) identified from (i) ?

Identification of Interventional Direct/indirect Effects

Assumption (Separability)

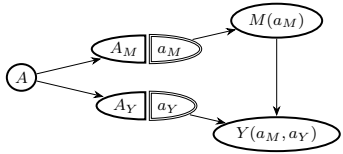
$$M(a_M, a_Y) = M(a_M)$$

$$P\{Y(a_M = 1, a_Y) \,|\, M(a_M = 1) = m\} = P\{Y(a_M = 0, a_Y) \,|\, M(a_M = 0) = m\}$$



- A_Y component does not causally affect M
- Imply no unmeasured confounder between M and Y

Identification of Interventional Direct/indirect Effects



Assumption (Separability)

$$M(a_M, a_Y) = M(a_M)$$
$$P\{Y(a_M = 1, a_Y) \,|\, M(a_M = 1) = m\} = P\{Y(a_M = 0, a_Y) \,|\, M(a_M = 0) = m\}$$

Theorem

Under the randomized A and the separability assumption:

$$P\{M(a_M = 0) = m, Y(a_M = 0, a_Y = 1) = y\} = P\{Y(a_M = 1, a_Y = 1) = y \,|\, M(a_M = 1) = m\} \times P\{M(a_M = 0) = m\}$$

- Therefore, $\mathrm{E}\{Y(a_M = 0, a_Y = 1)\}$ is identified by

$$\begin{aligned}
\mathrm{E}\{Y(a_M = 0, a_Y = 1)\} &= \sum_{m,y} \Big[y \times P\{M(a_M = 0) = m, Y(a_M = 0, a_Y = 1) = y\} \Big] \\
&= \sum_{m,y} \Big[y \times P\{Y(a_M = 1, a_Y = 1) = y \,|\, M(a_M = 1) = m\} \times P\{M(a_M = 0) = m\} \Big] \\
&= \sum_{m,y} \Big[y \times P\{Y = y \,|\, A = 1, M = m\} \times P\{M = m \,|\, A = 0\} \Big]
\end{aligned}$$

- This exactly matches Pearl's mediation formula (1)
- Their point estimates and SEs coincide with those of NDE/NIE (= use `mediation`!), though their interpretations differ

Testability of the Assumptions

Assumption (Separability)

$$M(a_M, a_Y) = M(a_M)$$

$$P\{Y(a_M = 1, a_Y) \,|\, M(a_M = 1) = m\} = P\{Y(a_M = 0, a_Y) \,|\, M(a_M = 0) = m\}$$

- In the future 4-arm trials, these assumptions can be tested

$$H_{0M}: \quad P(M|A_M = a_M, A_Y = 1) = P(M|A_M = a_M, A_Y = 0) \qquad \forall a_M \in \{0,1\}$$

$$H_{0Y}: \quad P(Y|M = m, A_Y = a_Y, A_M = 1) = P(Y|M = m, A_Y = a_Y, A_M = 0) \qquad \forall a_Y \in \{0,1\},\ \forall m \in \mathcal{M}$$

- This contrasts with the cross-world assumption in Pearl (2001)'s mediation framework, which cannot be verified using any randomized experiments

Outline

1. Overview
2. Principal Stratification
3. Pearl's Natural Direct/Indirect Effects
4. Robins and Greenland's Controlled Direct Effect
5. Treatment Decomposition Approach
6. Summary
7. Assignment

Comparison of Mediation Frameworks

- **Principal Stratification** (Frangakis and Rubin, 2002):
  - Conditions on $\mathcal{C} = \{M(1), M(0)\}$
  - Direct effect identified for always-takers and never-takers
  - Does not require that M is on the causal path
  - Identification is challenging (sometimes impossible)

- **Pearl's Mediation Analysis** (Pearl, 2001):
  - Enables decomposition: Total = Direct + Indirect
  - Requires cross-world independence, not testable in randomized experiments

- **Controlled Direct Effect** (Robins and Greenland, 1992):
  - Cleanly defined from $Y(z, m)$
  - No indirect effect defined (there exists controlled mediator effect)
  - Straightforward interpretation under joint intervention on A and M

- **Interventional Mediation Analysis** (Robins and Richardson, 2011):
  - Enables decomposition: Total = Direct + Indirect
  - Same identifying formula as in Pearl's mediation analysis
  - Requires a decomposition of A into A_M and A_Y, testable in future 4-arm experiments

Comparison of Mediation Frameworks

Framework	Direct Effect	Indirect Effect	Note
Frangakis and Rubin (2002)	Effect among NT/AT	-	Effect among CO/DF = Direct + Indirect Effects Need strong assumptions for point ID
Pearl (2001)	NDE	NIE	Require cross-world independence
Robins and Greenland (1992)	CDE	-	One can define controlled mediator effect
Robins and Richardson (2011)	IDE	IIE	Require treatment separability

Outline

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Assignment

- Analyze the dataset `causalweight::JC` (Job Corps) with the following variables:
  - $X =$ (`female, age, white, black, hispanic`)
  - $A =$ `assignment`
  - $M_1 = \mathbb{1}(\texttt{pworky4} > 0) =$ individual is employed in year 4
  - $M_2 = \mathbb{1}(\texttt{train1} + \texttt{train2} > 0) =$ participated in vocational training in year 1 or 2
  - $Y =$ `earny4`
- Goal: Estimate direct/indirect effects of JC assignment on 4th-year weekly earnings, mediated by M_1 or M_2
- Using the dataset, obtain appropriate effect estimates based on the following mediation frameworks:
  - Principal stratification (using M_1 as a mediator)
  - Pearl's mediation analysis (using M_2 as a mediator)
  - Controlled direct effect (using M_2 as a mediator)
  - Interventional mediation analysis (using M_2 as a mediator)
- In principal stratification, focus on the effect among always-takers, and explain:
  - Why we focus on this group
  - Whether the effect can be point-identified
  - If not, provide an appropriate alternative using partial identification
- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

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